

Research paper

Comprehensive study on social, compositional and thermal aspects of household solid waste for waste-to-energy potential estimation in Tashkent city

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ABSTRACT

The management of household solid waste (HSW) has emerged as a key issue in developing nations. A critical analysis of the physical and chemical components of MSW management and infrastructure is necessary to address current socioeconomic issues. According to the World Bank projections, the annual global generation of solid waste from municipalities is expected to reach 2.59 billion tons by 2030 and 3.40 billion tonnes by 2050. Inadequate handling of waste products frequently results in harm to the environment, the spread of diseases, and the generation of greenhouse gases, all of which have contributed to climate change and global warming. The waste-to-energy (WtE) strategy, which drastically reduces waste volume and produces renewable energy, has been adopted by a number of countries to address these issues. In this study, the social, compositional, and thermal aspects of household solid waste in a typical Tashkent District were studied. A Social survey was conducted to investigate residents' behavior towards waste sorting. The survey was conducted in the Uchtepa district of Tashkent, Uzbekistan. The composition of household solid waste was studied to explore the fractional distribution of the waste in the area. The American Society for Testing and Materials (5231–92) and European PN-EN standards were used for sampling, sorting, and for the rest of the analyses. Subsequently, proximate and ultimate analyses were performed to better understand combustion properties. A high heating value was identified to estimate the energy generation potential of household solid waste. Social survey results showed that the waste generated in the apartments will be discarded either on the day of its generation or the maximum after three days. Overall, 52 % of the respondents chose to sort their waste, compared to 48 % who had no real desire to separate their household solid waste. Food waste, plastics, and paper constituted 70 % of the total waste generated in the district. The mixed waste sample had a high heating value of 23.87 MJ/kg. The analysis revealed an energetic potential of HSW generated from the landfill as thermal energy.

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1. Introduction

The increase in total population, accompanied by rapid urbanization and industrialization, has led to a sharp increase in the volume of waste generated in modern societies. Increasing economic activity and food consumption, as well as changing lifestyles, are generating huge amounts of household waste, which poses a critical problem in developed and developing countries around the world. HSW management practices in developing economies face many challenges, including inadequate infrastructure and financial commitments that result in poor waste management services. Moreover, urbanization and population growth only serve to increase the amount of waste produced and the need for collection services in urban areas. Many developing nations lack credible information for adequate waste management, and those that do exist are frequently inconsistent, which makes it difficult to conduct a thorough and critical assessment of the various waste management strategies available for successful and efficient waste management programs (Jagaba et al., 2022).

Household solid waste poses various threats to public health and negatively impacts flora, fauna, and the environment, especially when it is not properly collected and disposed of. In addition, household solid waste releases toxic chemicals into the atmosphere and soil whenever it is decomposed or burned (Christine et al., 2014; Barinova et al., 2020). Environmental conservation problems are the most pressing problems in the modern world because the safety of the ecosystem is an important condition for sustainable development. According to statistics, only 5 % of the waste is recycled, and the remaining 95 % lies in landfills and occupies vast areas. HSW is typically surplus material from urban, rural, and industrial streams that has been rejected, abandoned, or disposed (Pop et al., 2022). Solid wastes can be liquid, semi-solid, or contain trapped gaseous materials, among other physical states. As urbanization picks up speed, the world faces a major challenge in maintaining a pollution-free planet because to the rapid increase of MSW. Globally, 2.01 billion metric tons of municipal solid waste are produced each year. An ecologically viable method of handling it is thought to be lacking in 33 % of cases, at most. In developing and economically disadvantaged countries, the percentage is considerably greater (Jerin et al., 2022).

The management of household solid waste is one of the most prevalent problems in the world. Waste is a serious threat to both the environment and public health, and has a negative influence on residents' day-to-day activities. Population expansion and an increase in waste production are factors that contribute to the need for efficient waste management. The daily generation of MSW varies depending on the region and level of economic growth, ranging from 0.11 to 4.54 kg. It is anticipated that waste creation will increase by 19 % and 40 %, respectively, in developed as well as industrialized countries (Kawai and Tasaki, 2016). With regard to HSW, current growth will be highest in South Asia and East Asia (Pacific) (Habib et al., 2021). The overall HSW generation in the South Asian region is predicted to increase by more than twice the current rate. In the Republic of Uzbekistan, resident generates roughly 0.6 kg/cap/day, or 220 kg/cap/year, of municipal solid waste daily. This indicator increases to 0.87 kg/cap/day or 300 kg/cap/year for each person in major cities. Approximately 7 million tons of MSW were produced in the country in 2017. Over 1.1 million tons, or 11 % of it, was produced in the city of Tashkent (Rakhmatullayev et al., 2020). Currently, in Tashkent, the waste generated daily is taken into two major landfills Okhangaron (489.29 hectare) and Tulabek (28.44 hectare). These landfill sites are approximately 100 km and 50 km from the city, respectively. The Tashkent City Department of Ecology and Environmental Protection under the Tashkent City Statistical Department stated that 1500–2100 tons of MSW were produced daily in Tashkent City in 2021. The number for the city was 744280 tons per year (TCSD, 2022). It should be noted that in 2021, the population of Tashkent was 2 694 400, according to the Department of Statistics under the City Administration.

While choosing the best collection strategy, treatment technology,

ultimate disposal, waste creation, and thorough characterization are among the most crucial elements to consider. Urbanization is causing developing nations such as Uzbekistan to see population growth and changes in lifestyle due to large-scale migration from rural to urban areas, such as the capital city Tashkent, in search of better economic opportunities. This has resulted in a significant increase in the production of solid waste from urban households. This condition places a lot of strain on landfills, particularly in large cities with a shortage of disposal areas. Thus, it is imperative to reduce and properly utilize solid waste.

Social surveys are described as a means of gathering quantitative as well as qualitative data from (normally representative) samples of people through their oral answers to uniform sets of systematic, structured questions presented either by interviewers or in self-completion questionnaires (Payne and Payne, 2004). Social surveys are best used to obtain information about large populations and measure public attitudes and orientations in a large population (Ali et al., 2022). Thus, social surveys help to understand public opinion about certain issues or contexts and establish necessary measures for the betterment of a situation by collecting data through questionnaires. In our case, it helped improve the Municipal Solid Waste (MSW) management system.

According to Sarbassov et al. (2019), integrating recycling services with environmental amenities may be beneficial for encouraging resident participation in Household Solid Waste (HSW) sorting and recycling. Moreover, they noted that various causes and circumstances may affect citizens' recycling and sorting habits. Obtaining the public involvement in solid waste management is crucial for the entire process.

Yildirim and others justify the necessity of surveys on MSW management as follows: Basic statistics on the types and quantities of food waste are based on limited data. This demonstrates the necessity of conducting more studies to enlarge an effective database. This finding emphasizes the value of gathering information on food loss and waste in the area (Yildirim et al., 2016). So, despite the fact that this research is based on a limited study, it is significant because, through an exploratory survey, it attempts to gather fundamental data on household behavior that enables extending the re-search based on many future-related characteristics (Yildirim et al., 2016). Before the implementation of recycling efforts, Sarbassov et al. (2019) claimed that interviewing individuals about their solid waste disposal practices and the possibility of sorting garbage at the source might provide useful information. They highlighted that the survey results are crucial for the formulation of new waste management plans based on source separation and for the improvement of resident participation in the existing HSW management systems in the region's capital city and other municipalities (Sarbassov et al., 2019).

Scientists from Kazakhstan examined how residents of Nur-Sultan (formerly known as Astana), the nation's capital, felt about the disposal and segregation of solid household trash at the source. With 3281 participants, the survey was carried out using stratified random sampling during the summer. This was the first time in Kazakhstan that recycling trends were evaluated from a household's point of view. The study also showed that separation at the source may be successful if local governments launch sensible public awareness programs and place recycling bins adjacent to residential structures (Sarbassov et al., 2019).

To better understand the problem of recycling solid waste and the advantages that accrue to families, a team of researchers from Nigeria conducted a sociological study in Kaduna City in the country's north-western region in 2016. Questionnaires were administered to a total of 500 homes by random selection. According to the findings, low-income families recycle more rubbish than high-income families do. The respondents' dwelling type also revealed information about their recycling practices. This has a strong impact on the city's overall environmental management (Abd'Razack et al., 2017). There is a strong link between educational awareness and waste-sorting behavior. Individuals with a formal education are better able to comprehend the significance of recycling and environmental conservation. According to the authors, there is a strong relationship between several

socioeconomic and demographic factors, including household size, recycling benefits, age of respondents, educational attainment, type of housing, and type of residence. However, there is little connection between respondents' recycling behaviors and their gender (Abd'Razack et al., 2017).

A rationalized environmental evaluation of waste management systems considers the chemical and physical composition of particular waste material fractions (Lange, 2021). Characterization is a method that helps with long-term municipal research and development as well as better waste management by creating a statistical portrait of the types and quantities of waste produced. The most appropriate disposal and/or treatment technique for recovering valuable resources trapped in a heterogeneous mixture of organic and inorganic components depends critically on the HSW composition. Numerous elements, including economic standing, past events, standard of life, energy use, and climatic change, among others, frequently influence the composition of HSW (Korai et al., 2017).

Recycling the organic components of HSW provides considerable management benefits, including environmentally sound waste management and recycling, which significantly reduces the environmental effects of solid waste management, including greenhouse gas emissions. Before the implementation of recycling efforts, interviewing residents about their solid waste disposal practices and the possibility of sorting waste at the source may provide important information. This study investigated whether it would be feasible to use household solid waste streams to produce green energy by physically characterizing the trash composition and conducting sociological and field surveys. In order to have current knowledge about the qualitative and quantitative features of settlements and their population, decision makers and entrepreneurs in the housing sector, spatial and infrastructure planning, municipal revenue collection, and service delivery can greatly benefit from the results of this study. Furthermore, determining the appropriate energy potential of a specific house solid waste source necessitates a thorough examination of the social, compositional, thermal, quantitative, and qualitative characteristics of the waste produced. In this sense, the purpose of this study is to assess social (source separation habits among residents), compositional (physical and chemical characteristics), and thermal aspects of HSW for waste-to-energy potential estimation.

2. Materials and methods

2.1. Social survey

Based on studies on HSW sorting, there are various causes and aspects that may affect citizens' recycling and sorting behaviors. In addition, literature analysis emphasizes the importance of including citizens in the handling of solid waste recycling as a crucial element of the entire process (Sarbasov et al., 2019; Raghu and Rodrigues, 2020). The views of locals on household solid waste in Tashkent City were examined through a social survey. Sarbasov et al. (2019) served as inspiration for this survey. Three essential results were obtained: frequency of HSW removal, source separation habits, and the most accumulated HSW categories. The survey methodologies are provided by outlining the research region and providing statistical information on the population. The findings of this study are useful for municipal governments in developing new waste management plans based on source-based separation and for enhancing resident participation in current HSW management systems.

2.1.1. Description of the study area

Tashkent is the capital and largest city in Uzbekistan, a city of republican subordination. It is also the largest city in Central Asia in terms of population (2 909 500 people in June 2022, according to the Tashkent City Statistical Department, 2022). Tashkent is located in the northeast of the country near the border with Kazakhstan. The city has 12 administrative districts, namely Almazar, Shaykhantakhur, Uchtepa,

Chilanzar, Sirgali, Yunusabad, Mirzo Ulugbek, Yashinabad, Yakkasaray, Mirabad, Bektemir and Yangiheyat which has only been established in 2020. The Administrative division and brief statistics of Tashkent City are given in Fig. 1.

The population of Tashkent increased from the south to north of the city. Almost half of the population (49.07 %) live in the northern districts of the city (Shaikhontohur, Almazar, Yunusabad and Mirzo Ulugbek). This population lives in an area equal to over one-third (36.8 %) of the entire city area. Approximately, 30 % of the population lives in the western (Uchtepa and Chilonzor) and eastern (Yashinabad) districts, and these districts occupy 26.76 % of the total area of the city. The remaining 20 % of the population lives in the southern districts of the city, which occupy about 36.41 % of the territory.

The demographic profile of Tashkent City (Table 1) shows an average of 69 % of the total population has been the age group of 16–69 which is physically active and healthy part of society. Following that, the 0–15 age group had an average 25.5 % share in total. Lastly, the 70 and older age groups had an average 4.6 % share in total. As sorting habits depend on age, it is important to study the distribution of different age groups in Tashkent City. Looking at the trends, the 0–15 and 70 and older age groups showed a positive trend between 2015 and 2022. While, 16–69 age group was negative during the given period.

From Fig. 2, the population trends can be studied in each region in the given years, and comparisons can be made. As studied above, Yunusabad, Mirzo Ulugbek, Shaykhontokhur and Almazar districts had the highest population among others in 2010. Steady growth had all four of these districts in 2015, 2020, and 2022, apart from a significant shift in Mirzo Ulugbek between 2020 and 2022. Following these four districts, Uchtepa, Chilanzar, and Yashinabad have relatively high populations compared to other districts that have not been described. Regarding trends, the population rose steadily in all three districts in the given years. However, from 2020 to 2022, the population in Yashinabad rose sharply, left behind in the Chilanzar, and nearly reached the population in Uchtepa. The other districts had minor increases and decreases during the given years. The Yangiheyat district was established in 2020; thus, by adding more area from the Tashkent region, the population will increase from 2022.

The percentage share of participation is shown in Fig. 3. In this case, a multi-stored residential building was selected as the research object. There were 80 apartments in the 5-story building 33 of these households participated in the questionnaire, and 47 could not participate. Of these 47 households, 36 did not participate because no one was at home and eight because no one lived in the household.

2.1.2. Design of survey

Most high-rise buildings are common in cities. Naturally, most of the population lives in these houses. Therefore, household waste refers to waste generated in multi-stored buildings. It does not include business centers, offices, government agencies, and business entities. As a matter of fact, they have a share in total waste, but since household waste was studied in our work, the above-mentioned sources were not studied. For this reason, a social survey was conducted among residents living in apartment buildings.

This study aimed to evaluate several important factors such as source separation habits among Tashkent citizens. The survey was conducted in November 2021 in the Uchtepa 23rd District. This area was chosen as a representative of an ordinary bloc in Tashkent city with approximately 20,000 residents and covers 0.65 km². The flowchart of the survey is shown in Fig. 4.

First, general information of the study area was investigated, such as residential and non-residential buildings, waste collection points, and most importantly, the overall population. The question form was then prepared with five questions that were considered essential for the survey. These five questions covered waste removal and sorting attitudes of the residents, main types of HSW generated in the house, approximate mass of generated waste ready for removal, and the

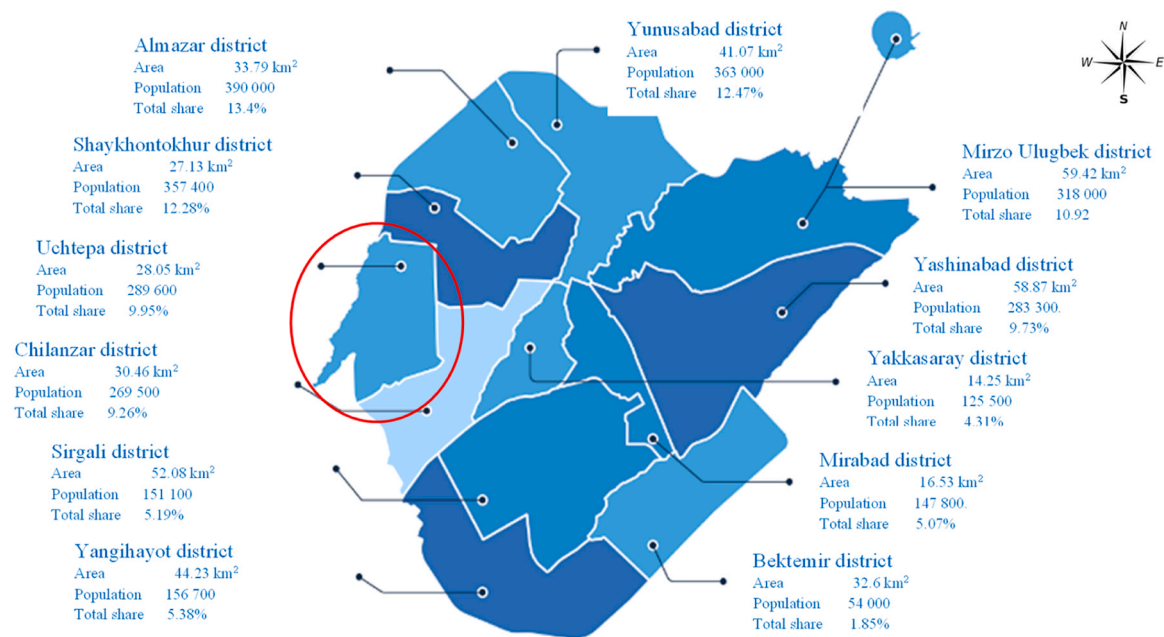


Fig. 1. Map of Tashkent City administrative division with area and population data in 2022, June (TCSD, 2022).

Table 1
Demographic profile of Tashkent City between 2015 and 2022 based on (TCSD, 2022).

Years	Age distribution						Total	
	0–15		16–69		70 and higher			
2015	592368	24.96 %	1673337	70.51 %	105564	4.45 %	2373284	100 %
2016	601554	25.12 %	1686612	70.42 %	105010	4.38 %	2395192	100 %
2017	615705	25.38 %	1699755	70.06 %	108686	4.47 %	2426163	100 %
2018	634375	25.71 %	1716835	69.59 %	113723	4.6 %	2466951	100 %
2019	653999	26.03 %	1737528	69.16 %	118442	4.71 %	2511988	100 %
2020	682005	26.49 %	1764281	68.55 %	125382	4.87 %	2573688	100 %
2021	727702	26.98 %	1835467	68.07 %	131209	4.86 %	2696399	100 %
2022	780635	27.25 %	1942697	67.82 %	139088	4.85 %	2864442	100%

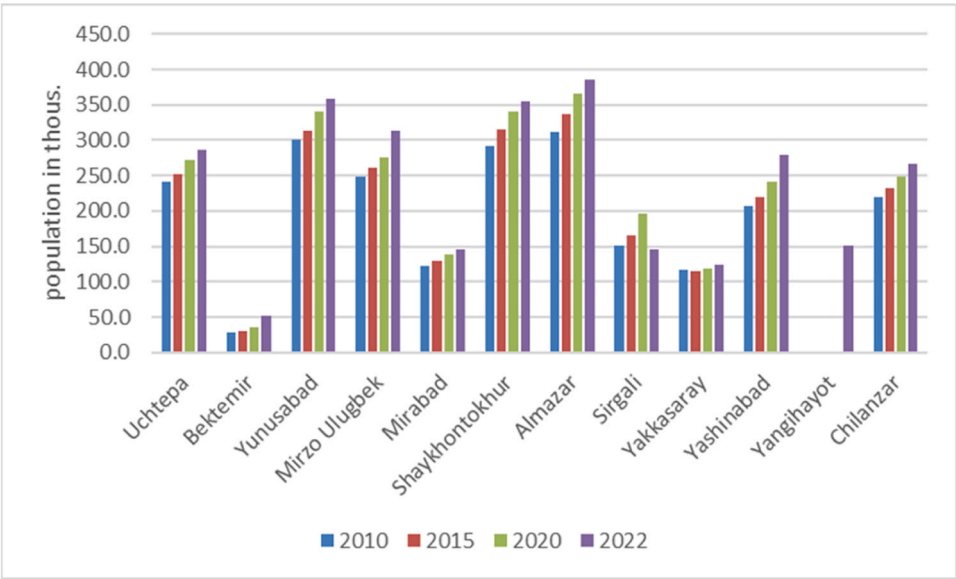


Fig. 2. Population distribution with latest trends from 2010 to 2022 by districts (TCSD, 2022).

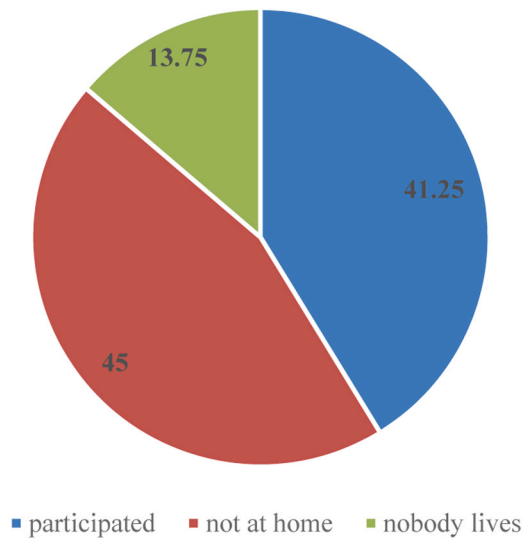


Fig. 3. Overall view of participation.

number of people living in the house. Once the form was ready, the questionnaire was printed based on the expected number of participants. Subsequently, the survey object was identified based on the number of apartments, location, and distance from nearby Waste Collecting Points (WCP). Face-to-face interviews were conducted with each participant by visiting their apartments. The survey was conducted on Wednesday in the late afternoon, which was the appropriate time for visiting. In this investigation, commercial buildings, schools, and other non-residential buildings were excluded because the survey focused on residential areas and waste generated from there (Sarbasov et al., 2019). The interviews were arranged such that each participant spent no more than five minutes. After self-introduction, five multiple-choice questions were asked and the answers were recorded. Finally, recommendations and feedback are noted as additional data.

2.2. HSW Sampling methodology

Solid waste composition or waste composition is described as the characterization of solid waste, as represented by a breakdown of the mixture into specified waste components on the basis of mass fraction or weight percent (ASTM, 2016). To investigate waste types and their share in household solid waste, a compositional study was carried out in the 23rd block of the Uchtepa district in Tashkent city. The study was

conducted during the autumn season. All waste collection points were identified, and two of them, which were close to the building social survey, were chosen for gathering waste samples. According to the American Society for Testing and Materials (ASTM) D 5231–92 standard, the sample weight is must be 91–136 kg in order to represent the characteristics of a vehicle load of MSW (ASTM, 2016). Thus, in this study, a 100 kg waste sample was extracted from the WCP. The samples were transferred from the bins to special bags and weighed using a scale.

2.3. HSW sorting procedure

A clean flat area was prepared as according to the ASTM procedure (ASTM, 2016), and the waste bags were unpacked. Normally, each portion of the HSW is placed in a separate plastic bag and there are many of them. Thus, waste must be unpacked from plastic bags to form a representative sample of HSW. The unpacked waste pile was then thoroughly mixed from the four sides using a shovel. Following this, using the quartering method, well-mixed waste piles were divided into four equal parts, and two diagonal sides were discarded. The other two sites were taken as representative samples of HSW for the sorting stage. In the sorting stage, samples were mixed for a second using a shovel, and the segregation process was performed. Based on the ASTM D 5231–92 waste composition data sheet (ASTM, 2016), seven main types of waste were sorted: paper, plastic, food waste, ferrous, glass, textile, and others. First, all plastics were segregated as they were abundant in the waste. Then, the paper, textile, glass, and metals were collected. On the final stage, food waste was separated which what was left was other types. The sorted waste fractions were placed in seven labelled special plastic bags, and the weight of each fraction was identified on a scale. Finally, the results were recorded on a waste composition data sheet. For laboratory analyses, a kilo of samples from each fraction was taken.

2.4. Proximate and ultimate analysis

Biomass contains a large number of complex organic compounds, moisture, and a small amount of inorganic impurities known as ash. Organic compounds consist of four principal elements: carbon (C), hydrogen (H), oxygen (O), and nitrogen (N). Biomass, such as MSW, may also contain small amounts of chlorine (Cl) and sulfur (S). Cl is rarely present in biomass, except for secondary sources such as demolished wood. A thermal design of a biomass utilization system, whether it is a gasifier or combustor, requires the composition of the fuel, as well as its energy content (Fantini, 2017). Proximate and ultimate analysis are mostly used in the context of thermal conversion, such as combustion (Basu, 2013; Onokwai et al., 2022; Fajobi et al., 2022; Chandraratne and

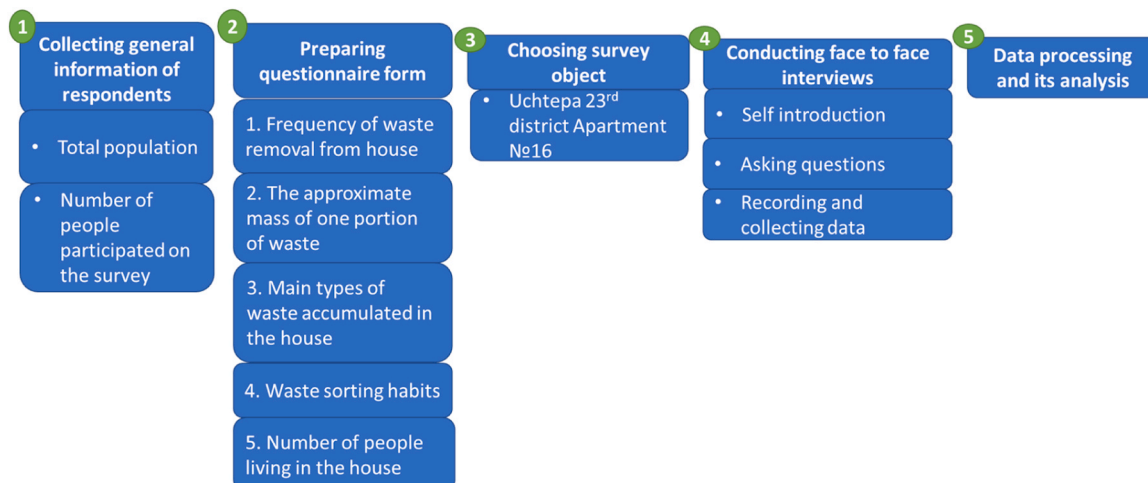


Fig. 4. Flowchart of the survey design.

Daful, 2022).

Proximate analysis was performed to identify the four main combustion parameters of each HSW sample: moisture, volatile matter, fixed carbon, and ash content. The analysis was performed at the Laboratory of Energy and Fuels of the Stanislaw Stashits Mining and Metallurgy Academy in Krakow, Poland in 2022. The Eltra TGA Thermostep analyzer was used in the laboratory experiment.

A high moisture content is a major characteristic of biomass. Moisture drains much of the deliverable energy from a gasification plant because the energy used in evaporation is not recovered. This important input parameter for design must be known to access the cost of transportation or the energy penalty in drying the biomass (Motta et al., 2018; Tuomi et al., 2019; Siddiqui et al., 2022). The moisture in biomass can remain in two forms: free, external, and inherent or in equilibrium.

Basis of Expressing Moisture: Biomass moisture is often expressed on a dry basis(db). For example, if W_{wet} kg of wet biomass becomes W_{dry} after drying, its db (M_{dry}) is expressed as:

$$M_{dry} = \frac{W_{wet} - W_{dry}}{W_{dry}} \quad (1)$$

This can provide a moisture percentage greater than 100% for very wet biomass, which might be confusing. Therefore, the basis of moisture should always be specified. The wet basis (M_{wet}) and the dry basis (M_{dry}) are related as:

$$M_{wet} = \frac{W_{wet} - W_{dry}}{W_{wet}} \quad (2)$$

$$M_{dry} = \frac{M_{wet}}{1 - M_{wet}} \quad (3)$$

The volatile matter of a fuel is condensable, and noncondensable vapor is released when the fuel is heated. Its amount depends on the heating rate and temperature to which it is heated.

Ash is the inorganic solid residue that remains after the fuel is completely burned. Its primary ingredients are silica, aluminum, iron, and calcium, small amounts of magnesium, titanium, sodium, and potassium may also be present (Basu, 2013).

Fixed carbon in biomass is determined from the following equation:

$$FC = 1 - M - VM - ASH \quad (4)$$

where M, VM, and Ash represent moisture, volatile matter, and ash, respectively.

Fixed carbon represents the solid carbon in the biomass that remains in the char during pyrolysis after devolatilization. During the determination of volatile matter, a portion of organic carbon is transformed into a carbonaceous material called pyrolytic carbon. Because fixed carbon depends on the amount of volatile matter, it is not directly determined. The volatile matter also varies with the heating rate. In a real sense fixed carbon is not a fixed quantity, but its value, measured under standard conditions, provides a useful evaluation parameter for biomass (Basu, 2013; Suryawanshi et al., 2023).

Ultimate analysis is used to determine the C, H, N, S, and O content in HSW. Such analysis presents the weight percent of carbon, hydrogen, oxygen, nitrogen, and Sulphur in the sample simultaneously, and the weight percent of oxygen is determined by difference (Tursunov et al., 2015a,b). Ultimate analysis was done at the International Laboratory of Energy and Fuels of the AGH University of Science and Technology, Poland in 2022. The CHS-580 analyzer by Eltra was used in this analysis, and its working principle is given in Fig. 5 (Tursunov et al., 2015a,b).

The weight was transmitted from the interface balance to the computer once the sample was weighed in a ceramic boat. The ceramic boat was then placed in a furnace burning. A typical analysis took between 60 and 180 seconds. During the analysis, detector signals and instrument characteristics were displayed on the monitor. The signals were automatically evaluated, and the findings are shown. The data can be sent to a Laboratory Information Management System (LIMS). The chemicals and particle filters that must be maintained are readily available. The sample was burned in CHS-580 at temperatures as high as 1550 °C in an oxygen environment. The furnace temperature was adjusted at will in increments of 1 °C. The combustion gases (CO_2 , H_2O , and SO_2) were passed through a dust filter before entering the heated H_2O infrared cell. Dry CO_2 and SO_2 gases were discovered in the extra infrared cells following the chemical absorption of water vapor. Depending on the arrangement, up to four infrared cells (for CS analyzers) or up to three infrared cells (for CHS analyzers) with various sensitivities can be combined (Fig. 5).

Bases for Representing Biomass Composition: Depending on the context, several bases are frequently used to indicate the composition of a fuel. The following four analytical bases are frequently employed:

1. As-received
2. Air-dry
3. Total dry
4. Dry and ash-free.

A comparison of these bases is shown in Fig. 6.

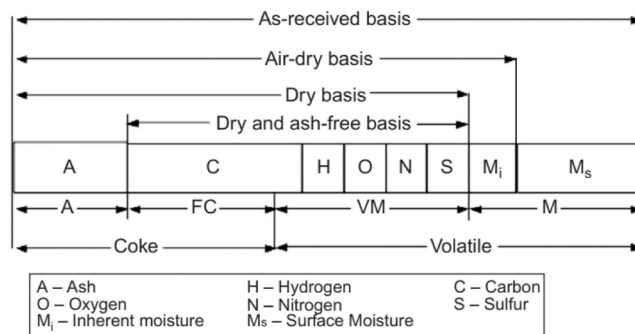


Fig. 6. Basis of expressing fuel composition.

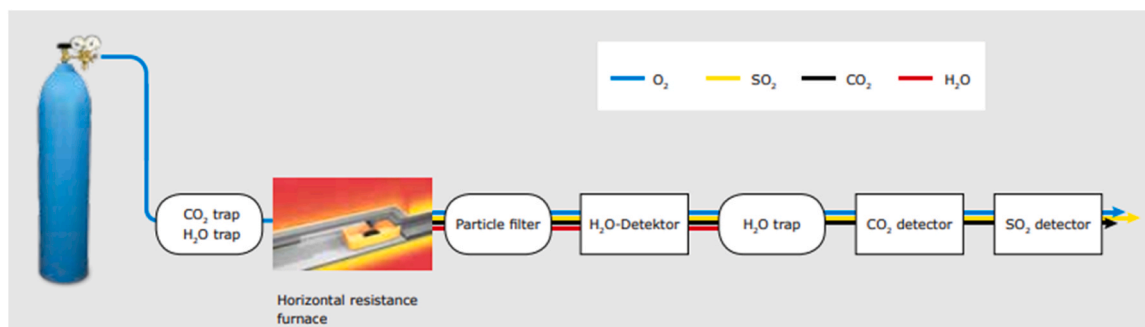


Fig. 5. The CHS-580 analyzer working principle (Tursunov et al., 2015a,b).

As-Received Basis:

When using the as-received (ar) basis, the results of ultimate and proximate analyses can be written as follows:

$$\text{Ultimate analysis : } C + H + O + N + S + \text{ASH} + M = 100\% \quad (5)$$

$$\text{Proximate analysis : } \text{VM} + \text{FC} + M + \text{ASH} = 100\% \quad (6)$$

Where C, H, O, N, and S represent the weight percentages of carbon, hydrogen, oxygen, nitrogen, and sulfur, respectively, as measured by ultimate analysis, and VM, FC, M, and ASH represent the weight percentages of volatile matter, fixed carbon, moisture, and ash, respectively, as measured by proximate analysis. Both studies reported the same amounts of ash and moisture in the fuel. The base "as received" can be changed to another base.

Basis for air-drying:

When fuel is dried in air, its underlying moisture is maintained but its surface moisture is removed. Hence, the quantity is divided by the total mass less the surface moisture in order to represent the component on an air-dry (ad) basis. For illustration, the carbon proportion on the "ad" basis was determined as follows:

$$C_{ad} = \frac{100C}{100 - M_i} \quad (7)$$

where M_j is the amount of surface moisture taken from 100 kg of fuel after drying in air. Similar expressions were possible for the other fuel component parts.

Total Dry Basis:

The fuel composition on the "ad" basis is a practical parameter that is easy to measure, but to express it on a totally moisture-free (td) basis, we must make an allowance for surface as well as inherent moisture, M_i . This gives the carbon percentage of the total db and C_{td} :

$$C_{td} = \frac{100C}{100 - M} \quad (8)$$

where M is the total moisture (surface + inherent) in the fuel: $M = M_s + M_i$

Dry Ash-Free Basis:

Ash is another component that is eliminated along with moisture at times. This provides the fuel composition on a d_{af} basis. Following the aforementioned examples, the carbon percentage on a "daf" basis, C_{daf} , can be determined.

$$C_{daf} = \frac{100C}{100 - M - \text{ASH}} \quad (9)$$

where $(100 - M - \text{ASH})$ is the mass of biomass without moisture and ash.

The percentage of all constituents on any basis totals 100. For example:

$$C_{daf} + H_{daf} + O_{daf} + N_{daf} + S_{daf} = 100\% \quad (10)$$

2.5. HHV/Calorific value

The amount of energy that biomass is released when it is entirely burned in sufficient oxygen is known as the heating value of the biomass. Energy conversion is one of the most crucial biomass characteristics. Because of its extremely low density and high oxygen content, biomass has a low heating value compared to the majority of fossil fuels (Chen et al., 2021; Fajobi et al., 2022), especially when measured on a volumetric basis (Basu, 2013).

Identification of the energetic value or heating value of HSW biomass samples can be achieved using mathematical models, structural and proximate analyses, and experimentally. Experimental determination using a bomb calorimeter requires a sample size of 1–2 g, which is inadequate to account for the vast variance in waste biomass composition, thus requiring a larger sample size (Kathiravale, 2003). Calorific

value analysis was performed at the Laboratory of Energy and Fuels, AGH University of Science and Technology in Krakow in 2022, Poland. Leco AC calorimeter was applied on this analysis

3. Results and discussions

3.1. Social survey

A social survey was conducted to investigate the residents' attitudes toward household solid waste. According to Fig. 7, the majority of respondents take out trash every day, which consists of 42 % of the total. The next time interval that collected waste was once in a three-day period, that is 32 % of the total respondents attended the questionnaire. Following this, every other day is the third most preferred time interval for people, with 23 % of people choosing this option. Finally, very few people push back to take out household waste into the waste collection point once a week. Only 3 % of the residents chose this option.

It is concluded that the waste generated in the apartments located in the research object will be removed at least on the day of its generation or maximum after three days. This means that every family disposes of its household waste at least twice a week or more.

Fig. 8 presents the waste sorting practices among residents in the Uchtepa 23rd district in Tashkent city in 2021. The graph indicates that the share of people who segregate their waste is 32 %, which is the highest share. During the survey, people who were segregated were mostly older people and new families. A similar result was observed in a survey conducted by Sarbassov et al. (2019). In the questionnaire, sorting means collecting organic, plastic, paper, and other waste types separately, which facilitates further treatment processes. Following this share, the opposite attitude comes with 29 % of the rarely segregating group. Next, people voted "often" and "never" options 20 % and 19 % respectively.

Overall, Fig. 8 highlights a general view of people who tend to sort their waste and those who have not. This can be simplified by grouping "Always" plus "Often" together and "Rarely" plus "Never" together. In total, this can be seen as 52 % of the residents have a wish to sort, while 48 % of them do not have strong wish to segregate their household solid waste.

Fig. 9 shows the types of waste generated in the household. The most frequent and common waste types were selected as options and votes were collected from each family. We also tried to make the questions and options easier for people to understand. As expected, food waste had the highest volume among all waste types. The second most frequent waste type was plastic, such as plastic bags and packaging. Next comes paper waste. The other three types, textile, metal, and others, were the least common fractions of household solid waste.

It can be concluded that food waste, plastic waste, and paper waste are the three most common waste types, and this should be proven with compositional study results. This is because of the consumption and eating habits of the residents. Based on this, food waste stands in the first place, following which plastics and paper waste are generated due to the packaging styles of products. Metals, textiles, glasses, and other waste

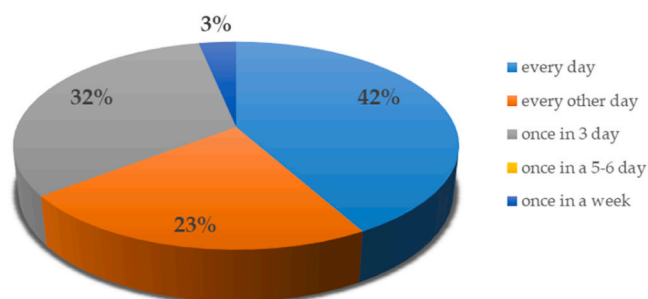


Fig. 7. Frequency of waste removal from residential houses.

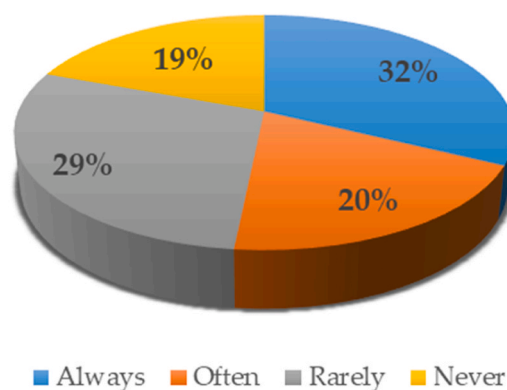


Fig. 8. Source separation behavior of the respondents.

types are rare because they exist in small amounts in people's daily life products. These waste types might be generated from kitchen utensils, electronic equipment, vases, etc. This type of product is not easily thrown away, and takes longer than organic or plastic wastes.

3.2. Composition of HSW

A compositional study of the HSW was conducted at selected waste collection points. The procedures were performed based on the research work of Tursunov and Abduganiev (2020), and sampling was performed based on the ASTM D 5231 standard. The values and results are presented in Fig. 10.

Overall, Food waste, plastic, paper, and other types are the top four wastes among household waste fractions with shares of 59 % or 30.875 kg, 11 % or 5.695 kg, 7 % or 3.78 kg and 13 % or 6.605 kg, respectively. Among these, food waste constitutes nearly 60 percent, making it dominant in the overall HSW. This is because of its continuous consumption and high moisture content, which provides a high volume and mass.

This composition helps us estimate the approximate amount of food waste, which is an energy resource in our research. Thus, Fig. 10, given above, clarifies that if household waste is considered a resource for Anaerobic Digestion (Slorach et al., 2019; Meng et al., 2022), approximately 60 % of the total waste will be available for the technology as a

feedstock.

Fig. 2 was prepared based on the sources given above. Various studies on the HSW composition and their results have been reported. It can be concluded that food waste constitutes 59.02 % of household waste. With this, we can calculate the approximate amount of food waste used for energy generation.

Thus, based on the data given for Tashkent City in 2021, which is 744280 tons per year, the annual mass of food waste generated in 2021 could be 446580 tons (about 60 % from the total amount). The results of the different compositional studies are presented in Table 2. It is clear that food waste, plastics, paper, and textiles generate the majority of the solid waste. However, it varies depending on where the waste samples are collected. Landfill samples, garbage containers, and waste collection points show different results. Thus, each of these results can be accepted depending on the scale of research.

3.3. Proximate and Ultimate analyses

Proximate analysis consists of four analyses: moisture content, volatile matter content, fixed carbon, and ash content determination (Tursunov et al., 2015a,b). The moisture content indicates the share of water in the biomass, which is represented as a percentage of the raw material weight. A high moisture content results in a notably lower carbon burning rate (Cavalaglio et al., 2020), as cited by Grandesso et al. (2011). Moisture decreases the heating value: the volatile matter or vaporous part of biomass is formed by condensable vapor and permanent gases (except for water vapor) emitted when the biomass is heated to 925 °C for a few minutes. Throughout this heating process, the biomass broke down into gases, and the solid matter was left out as char. The existence of volatile matter in biomass impacts fuel reactivity, and it has been investigated that increasing the content of volatile matter in the biomass sample causes, as a general tendency, a rise in the peak temperature. The peak temperature is the moment on the burning profile at which the rate of mass loss owing to combustion is at its maximum. The burning profile peak temperature is normally accepted as a parameter of the sample reactivity (Cavalaglio et al., 2020), as cited by Haykırı-Açma (2003). The ash content characterizes the volume of the solid and not the organic residue left after full combustion of the biomass (Cavalaglio et al., 2020). Fixed carbon is a solid burnable residue that remains after the biomass is heated and volatile matter is released, except for ash and moisture content (Cai et al., 2017;

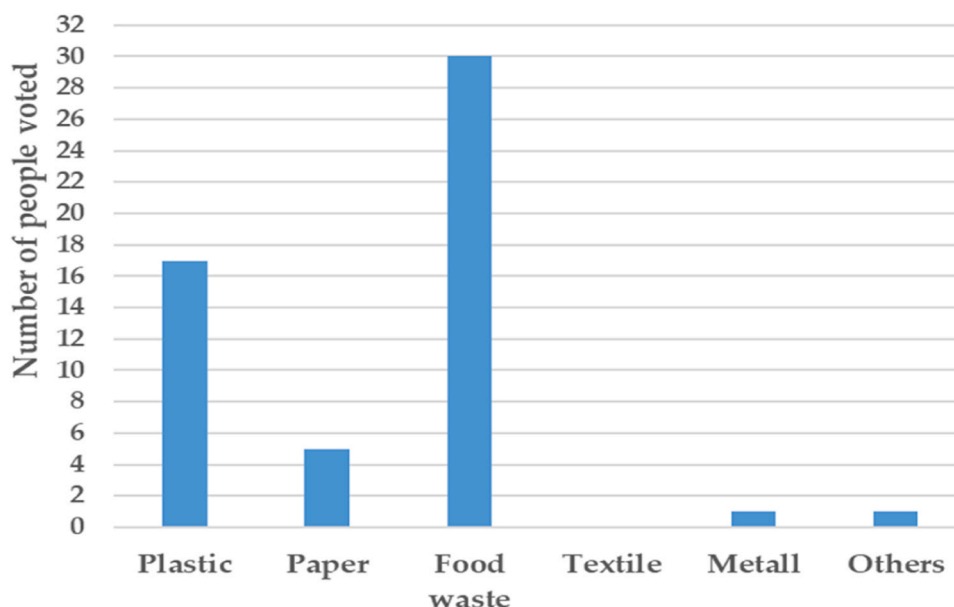


Fig. 9. Waste generation in the household by types.

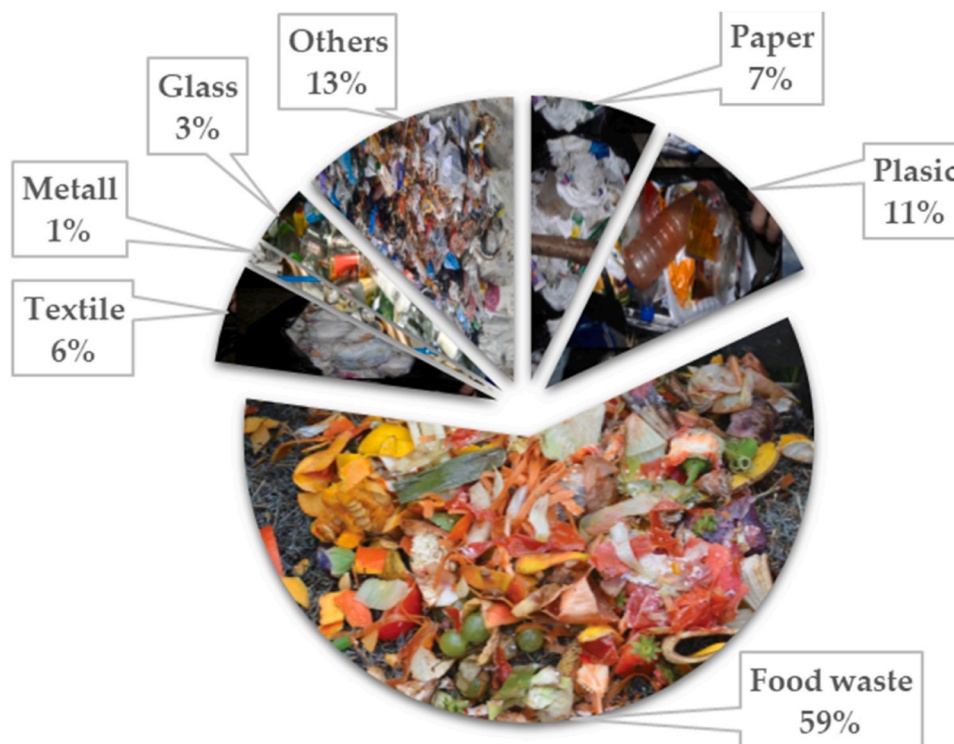


Fig. 10. MSW composition in the Uchtepa district of Tashkent city.

Table 2

Various HSW compositional studies conducted in Uzbekistan in different years.

Samples taken from	-	unknown	Urta-chirchik Landfill site	Garbage containers	WCP
Publication year	2008	2018	2020	2020	2022
Actual data experiment conducted	-	-	2019, January	2017–2018	2021, December
Food waste, %	38	2.5	9.5	53.42	59.04
Plastics, %	3	7.91	14.2	5.91	10.89
Paper and cardboard, %	25	3.22	4.7	3.59	7.22
Textiles, %	3	3.28	13	1.65	5.5
Rubber and leather, %	1.5	2.29	2.4	0.53	-
Metals, %	1.5	1.38	0.6	1.28	1.25
Glass, %	4	4.62	5.3	3.5	3.38
Wood, plant waste, %	6	14.3	47.9	2.66	-
Others, %	18	35.44	2.4	27.46	12.63
Reference	(Azizov, 2008)	(Open Data Portal, 2020)	(Tursunov and Abduganiev, 2020)	(Rakhmatullayev et al., 2020)	Current study

Cavalaglio et al., 2020).

The proximate analysis results are presented in Table 3. The experiment was performed three times and the average was considered as the final result.

It is clear from the table that, food waste had the highest moisture content of 41.24 %. This is because of the heterogeneous nature of food

Table 3

Proximate and Ultimate analysis results.

Order	Analysis, wt%	Plastic	Food waste	Textile	Mixed waste
<i>Proximate analysis</i>					
1	Moisture	0.94	41.24	3.51	35.96
2	Volatile matter	95.5	36.84	82.26	46.62
3	Fixed carbon	1.59	16.61	7.22	10.29
4	Ash	1.97	5.31	7.01	7.13
<i>Ultimate analysis</i>					
1	Carbon	79.5422	25.8168	45.2355	28.9875
2	Hydrogen	13.5442	16.0271	8.3417	15.6201
3	Nitrogen	0.0477	2.0231	0.0439	1.6871
4	Sulphur	0.0383	0.1950	0.0528	0.1703
5	Oxygen	6.8276	55.938	46.3261	53.535

waste, which is suitable for Anaerobic Digestion (AD) technology. After food waste, the second highest moisture content was found in the mixed waste.

According to proximate analysis, textile and plastic wastes have the highest volatile matter contents of 82.26 and 95.5 %, respectively, which means that they tend to vaporize well during burning. As stated in other studies, food and mixed wastes had the highest moisture content (41.24 and 35.96 %). The HHV of HSW will decrease and heat loss will rise with high moisture content (Yan et al., 2022; Manjunatha et al., 2020; Bhatsada et al., 2023). Higher levels of moisture facilitate the transfer of heat through radiation and, to some extent, facilitate the binding of particulates. Given that researchers have demonstrated a direct correlation between the carbon content of HSW and fuel quality, it can be inferred that converting this waste fuel into energy through an environmentally friendly thermochemical conversion process can result in a sufficient amount of energy. Regarding fixed carbon, mixed waste and food waste had higher values than the others (10.29 and 16.61 %). Finally, the ash content, which is a residue or leftover of the burnt material, was higher in the textile and mixed waste (7.01 and 7.13 %, respectively).

The percentage share of each individual element present in a material is explained by its total elemental analysis or ultimate analysis. The ultimate analysis results were used to describe the chemical composition of the organic fractions of the HSW samples. This analysis is important for evaluating the suitability of waste as a fuel and makes it possible to predict the emissions from combustion. The analysis results were also applied to define the appropriate mixing ratios of HSW materials to achieve suitable nutrient proportions (e.g., C/N) for biological conversion purposes such as composting (Pichtel, 2010). These elements are the main constituents of biomass and identify fuel potency and possible contaminating actions (Cavalaglio et al., 2020). The experiment was conducted three times and the average is considered as a final result.

The results from elemental analysis showed that food waste had the lowest carbon content compared to the other waste samples while highest content was observed for the rest of the elements (C: 25.8 %, H: 16 %, S: 0.19 %, N: 2 %, and O: 56 %). This means that when food waste is used for pyrolysis, it produces more SO_x-and NO_x contained gases than other waste types (e.g. textile, mixed waste, plastic). Because of the high oxygen content, the HHV of food waste is expected to be lower than that of other waste types, as demonstrated by energy estimation analysis. In addition, a lower carbon content indicated a lower HHV. In terms of carbon content, plastic waste had the highest percentage, which led to a higher energy content when combusted.

Hydrogen (H) has a negative influence on High Heating Value (HHV). This is due to the fact that hydrogen is associated with the water content (Tursunov and Abduganiev, 2020). It is known that the moisture content reduces the HHV of biomass. As a result, hydrogen shows the opposite proportionality to the heat derived from combustion. In the experiment conducted by Tursunov and Abduganiev (2020), the proportion of sulfur (S) and nitrogen (N) in wood waste samples was smaller than that in HSW, which they explained as follows: during thermochemical transformation, sulfur and nitrogen emissions from wood waste were significantly lower than those from HSW. Carbon content increased the heating value of the biomass sources. Some of the carbon is in the volatile matter, and another fraction forms most of the char (Cheng et al., 2020; Shah et al., 2023; Dada et al., 2023). Hydrogen also contributes to the heating value of the fuel, and during pyrolysis, most of the hydrogen is present in volatile matter. In contrast, oxygen content reduces the heating value (Tumuluru et al., 2021). The nitrogen content of fuel is a significant source of NO_x pollutants. A high concentration of fuel, such as agricultural and horticultural waste, leads to higher NO_x concentrations in the flue gases. Sulfur is a source of SO_x. Water in combustion gases leads to the formation of HCl rather than Cl₂ (Rose-ndahl, 2013).

3.4. HHV/Calorific value

The high heating value of any fuel is the amount of energy released per unit mass per unit volume of fuel when it is entirely burned out. The HHV indicates the condition in which water is evaporated from the combustion products and represents the gross calorific value (Cavalaglio et al., 2020). The identification of the energetic value or heating value of HSW biomass samples can be studied using mathematical models, structural and proximate analyses, and lab experiments (Tursunov and Abduganiev, 2020).

The heating value of a solid fuel, often known as the calorific value or heat of combustion, is the conventional indicator of the fuel's energy content. The energy content estimation results in Table 4 show that

plastic waste had the highest heating value of 47.17 MJ/kg. The second and the third the highest values of 24.39 and 23.87 MJ/kg recorded by mixed waste and food waste samples. The textile exhibited the lowest heating energy content of 18.63 MJ/kg. It is noticeable that food waste and mixed waste have nearly the same heating value of 24 MJ/kg, which means that food waste can be used for pyrolysis technology in a mixed form with the other waste types. Thus, food waste can be used as feedstock for pyrolysis either separately or in a mixed form. The theoretical results obtained using Dulong's formula showed small differences with a maximum error of up to 9 %. It is proven that both results are nearly identical. Physicochemical analyses and energetic values allow us to further assess the actual energy generation potential and, hence, make it possible to conduct feasibility studies in the future. According to the calculation, textile waste has 18.88 MJ/kg, mixed waste has 22.51 MJ/kg, plastic waste has 44.92 MJ/kg, and food waste were 21.6 MJ/kg.

According to the investigation of Sukarni (2016), Malang city's MWS had a high HHV of 25.78 MJ/kg. This HHV is slightly lower than bituminous coal, which has an HHV of 27.8 MJ/kg, but it is higher than common biomass, such as wood, coconut shell, and rice husk, which have HHVs of 20, 20, and 15 MJ/kg. Assuming that the city of Tashkent produces a minimum of 1500 tons of waste per day, and according to our analysis, 83 % (1245,000 kg) of mixed waste (food waste 59 %, plastic 11 %, paper 7 % and textile 6 %) from the total waste can be combustible that could be used to generate energy roughly 28,024,950 MJ/day. According to this finding, HSW in Tashkent city emerges as a potent candidate to be utilized as solid fuel in the production of electricity.

4. Conclusion

The social survey provided the following results, which are true for the study object: the waste generated in the apartments located in the research object will be taken out at least the day of its generation or maximum after three days. Every family disposes of household waste at least twice per week. In total, 52 % of the residents have a wish to sort, while 48 % of them do not have strong wish to segregate their HSW. Food waste, plastic waste, and paper waste were the three most common types of waste generated in the study.

This compositional study was conducted as follows. If household waste is considered a resource for AD, approximately 60 % of the total waste will be available for use as feedstock. If pyrolysis technology is used, approximately 84 % of HSW can be used as feedstock.

Proximate analysis results revealed that food waste samples taken from the study object had 41.24 % moisture, 36.84 % volatile matter, 16.61 % fixed carbon and 5.31 % ash. Based on previous studies, volatile matter and fixed carbon content are essential indicators of the fuel properties of a material, which in our study was 53.45 %.

Ultimate analysis results: food waste samples taken from the study object have 25.8 % carbon, 16 % hydrogen, 0.19 % sulphur, 2.03 % nitrogen and 55.9 % oxygen.

Calorific value estimation results: According to laboratory analysis, food waste has 23.87 MJ/kg a heating energy. When the theoretical calculation method using Dulong's formula was used, this value was 21.6 MJ/kg.

HSW has the potential to be a valuable renewable energy source for Uzbekistan if appropriate waste management technologies (such as thermochemical treatments and biochemical treatments) and technical policies are introduced. The results of this study can be used as a starting point for future experimental waste research as well as by policymakers

Table 4
Comparison of HHV: experimental and theoretical.

Order	Analysis type	Equipment/Model	Plastic	Food waste	Textile	Mixed waste
1	Laboratory analysis, MJ/kg	LECO AC calorimeter	47.17	23.87	18.63	24.39
2	Theoretical calculation by Dulong's equation, MJ/kg	$HHV = 337C + 1428\left(H - \frac{O}{8}\right) + 9S$	44.928	21.6037	18.8875	22.5198

in developing waste management strategies for Uzbekistan's large cities. As this study has empirically demonstrated for a single city, the energy potential of waste should also be thoroughly and scientifically investigated on an experimental basis.

Author Contributions

Conceptualization, Obid Tursunov and Islom Karimov; methodology, Islom Karimov, Obid Tursunov and Katarzyna Śpiwak; validation, Obid Tursunov, Xun Hu and Yuguang Zhou; formal analysis, Katarzyna Śpiwak; investigation, Islom Karimov; data curation, Islom Karimov and Katarzyna Śpiwak; writing—original draft preparation, Islom Karimov, Obid Tursunov and Roman Uvarov; writing—review and editing, Obid Tursunov, Xun Hu and Yuguang Zhou; visualization, Alexander Kustov, Umi Fazara Md Ali and Roman Uvarov; supervision, Obid Tursunov. All authors have read and agreed to the published version of the manuscript.

Ethical Approval

Not applicable

Consent to Participate

Not applicable

Consent to Publish

All authors gave their consent to publish this manuscript

CRediT authorship contribution statement

Roman Uvarov: Writing – original draft, Validation. **Katarzyna Śpiwak:** Validation, Methodology, Data curation. **Islom Karimov:** Writing – original draft, Methodology, Conceptualization. **Obid Tursunov:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Umi Fazara Md Ali:** Visualization. **Alexander Kustov:** Visualization. **Yuguang Zhou:** Writing – review & editing, Validation. **Xun Hu:** Writing – review & editing, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

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